A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Unburnt fuel in vehicle exhaust gases **1** Evaporation from fuel tanks / spillage at filling stations **1** Solvents (paints, cleaning products) **1** Aerosol propellants **1** Gas leaks / fossil fuel exploitation (methane) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank **1** Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.3	**1 mark each** 1 mark for each of the 3 points below. **1** A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe **1** Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) **1** NOx are chemically reduced to nitrogen and oxygen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues **1** Brain damage / death if enough haemoglobin is inactivated – low doses worsen chronic heart disease **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** B (about 40:40:20 sand:silt:clay) **1** Balanced mix gives good drainage and aeration from the sand fraction **1** Water retention and nutrient (ion) retention from the clay fraction – easy root penetration and cultivation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **loam** → a balanced mixture of sand, silt and clay	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) **1** Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand **1** With only 5% clay there is little surface to hold ions so nutrients are leached – low water retention too **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeability** → how easily water passes through / lets water through • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** C (clay) holds much water so has a high thermal capacity: warms and cools slowly – A warms rapidly after cold weather **1** Clay particles held together by capillary water so roots find it hard to push between them – penetration is easy in sandy A **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Shake a known mass of soil with water in a measuring cylinder and leave to settle – particles settle in order of size (sand first, then silt, then clay; organic matter floats) **1** Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) **1** Radon travels up cracks in the rock and through foundations into buildings **1** It becomes concentrated inside enclosed / poorly ventilated buildings – control by ventilating foundations / sumps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16 + 1.5 = 17.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
03.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Where they live (geology / radon, altitude) **1** Occupation (radiographer, nuclear worker, aircrew, miner) **1** Medical history (number of X-rays / CT scans) **1** Lifestyle / diet (eg shellfish near discharges) – smoking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	2
03.4	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it – risks can be reduced (shielding, dose limits) **1** Symptoms of chronic low-dose exposure take a long time to develop **1** The same symptoms (cancers) have other causes so effects are hard to attribute **1** Accurate data on human exposure / doses is often unavailable (especially after accidents) **1** People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions • **Accurate** → close to the true value	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 2 points below. **1** The scatter / variability of results around each mean – about 68% of values lie within one SD of the mean **1** The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean	2
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** t-test **1** Comparing two means of measured (continuous) data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **t-test** → a test for a difference between the means of two normally distributed sets **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** There is a 30% probability that the difference occurred by random chance **1** Above the 0.05 (5%) level so the difference is not significant – the null hypothesis (no difference) is accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Two correlated variables may both be caused by a third variable that was not investigated **1** Significance only shows the result is unlikely to be chance – a mechanism / controlled experiment is needed for causation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **correlation** → a relationship in which the two variables change together • **causal** → one variable actually causing the change in the other **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test	2
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Increase the number of replicates (reduces the effect of variability) **1** Standardise other variables (soil, water, light) more closely to reduce scatter **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Cost increases as the percentage prevented increases **1 The relationship is not linear:** cost rises slowly at first then very steeply / exponentially above about 80-90% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = 11.0 (£ million) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 4 points below. **1 ALARA:** emissions should be As Low As Reasonably Achievable **1** It is rarely practical / affordable to reduce emissions to zero – low-level emissions may have acceptable impacts so an acceptable emission level is set **1 BATNEEC:** Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive **1** The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
05.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Industry / city may have expanded so there are more pollutant sources and total emissions have risen **1** Better understanding of the health / environmental effects so stricter standards are set **1** Improved technology becomes available (BATNEEC changes) / cost of better control falls **1** Pollutant may have accumulated (persistent) so background levels are higher **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
07.1	**1 mark each** 1 mark for each of the 5 points below. **1** No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) **1** Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas **1** Small scale matches local demand – electricity used locally without long transmission lines / grid connection **1** Low construction cost and simple technology maintained by the community **1** Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed – Kaplan turbines with adjustable blades cope with variable flow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	5
07.2	1 mark for the point below. **1** Lack of capital / finance, technical knowledge or grid access – seasonal drying of rivers – conflicts with irrigation water – planning / land ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
07.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Output is proportional to flow x efficiency: 100 x 0.10 x (40 / 90) **1** = 4.4 kW (accept 4.4-4.5) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kW) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
07.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost – high flows that clear gravel spawning beds lost – floodplain no longer flooded **1** Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis **1** Sediment trapped so less nutrient replenishment downstream – barrier to migrating fish (salmon, sturgeon) and isolation of populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** Most energy comes from non-renewable fossil fuels whose reserves are depleting – the rate of new discoveries has fallen sharply since the 1980s **1 Use causes unacceptable environmental damage:** atmospheric pollution, climate change, oil pollution, habitat loss **1 Economic / social:** depleting resources become more expensive, reducing future affluence – risk of an energy gap if new technologies are not developed in time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coal mining:** habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise **1 Oil and gas extraction:** spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO2 / CO2 / smoke **1** Pipelines causing habitat damage during construction – tanker transport risks **1** Fuel processing (refineries, coal washing) pollution – embodied energy of equipment and site development for all resources **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	4
08.3	**1 mark each** 1 mark for each of the 3 points below. **1** New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) **1** Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise **1** By then depletion would reduce supply faster than replacements could be built, causing shortages **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
09.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up **1** High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) **1** Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) **1** Birds of prey are at the top of the food chain so receive the highest concentrations **1** Sub-lethal doses caused thin eggshells that broke / reduced breeding success – higher doses caused sterility or death **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **trophic level** → the feeding level in a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly **1 Disadvantage:** high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase) – chronic effects (memory loss, depression) – suspected carcinogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	2
09.3	1 mark for the point below. **1** Pyrethroids are highly toxic to fish (even though mammal toxicity is low) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
09.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) **1** They are moderately persistent so remain in the soil long enough to be absorbed **1** Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats **1** So bees / pollinators are less exposed than when the crop is sprayed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Carbon:** combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere – deforestation reducing photosynthesis – ploughing / drainage increasing aerobic decomposition – new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane – marine pollution reducing phytoplankton – consequences: rising CO2 and CH4, climate change, ocean acidification, reduced soil organic matter • **Nitrogen:** Haber process doubling fixation – legume cultivation – drainage increasing nitrification and reducing denitrification – combustion NOx – sewage and fertiliser leaching – consequences: eutrophication, smogs, acid rain, N2O warming, NOx toxicity • **Counter-strategies carbon:** renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • **Counter-strategies nitrogen:** reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • **Evaluation:** some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS) – scale of fossil fuel dependence – feedback and time lags – link to circular economy and low-temperature natural processes – judgement	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas • **leaching** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NOx, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NOx, biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NOx).	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
10.2	1 mark for the point below. **1** The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Attributes:** accuracy, precision, representative samples, reliability, anomalies, validity – hypothesis and null hypothesis – preliminary studies • **Sampling design:** random vs systematic location and avoiding bias – timing (diurnal, seasonal, long-term, weather) – sample size and number and the running mean – standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • **Statistics:** variability, standard deviation and error bars, significance and p-values, correlation vs causation – choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • **Application to Paper 1 topics:** monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	

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